The per-voice Fourier signature: the bass is the spectrally purest voice

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Abstract

For three centuries music theory has held that the bass is special—the *basse fondamentale* of Rameau, the figured-bass tradition, Riemann's harmonic functions: the lowest voice carries the harmonic roots, the foundation. That has always been a *structural* claim. The discrete Fourier transform of pitch-class distributions (Lewin, Quinn, Amiot, Yust) turned chord "qualities" into numbers—the magnitude $|a_k|$ of the k-th coefficient measures a character: $|a_5|$ diatonicity, $|a_1|$ chromaticity, $|a_6|$ whole-toneness. That program reads a piece by *time*; it has never asked whether each *voice* carries a different spectral fingerprint. We do. Over 334 four-part Bach chorales the naive guess—that the fifth-walking bass is the most diatonic ($|a_5|$) voice—is *false* (the soprano is). But the data shows something sharper: the **bass is the spectrally purest voice**, lowest in chromaticity ($\alpha_1 = |a_1|/|a_0|$, paired $t = -20$) and whole-toneness (α_6, $t = -15$), *despite using the widest pitch-class vocabulary*—so this is not a cardinality artifact. The cause is **root realization** acting through two channels: functional roots move by fifths/fourths (odd intervals), which balance even/odd pitch-class weight and drive $\alpha_6 \to 0$ (the *parity* channel), and the bass's wide tessitura spreads its weight around the chromatic circle, lowering α_1 (the *dispersion* channel). The effect is **texture-bounded**: strong in functional-bass homophony (Bach chorales; Palestrina), it fades in imitative (Monteverdi madrigals) and pre-functional (Trecento) textures—so the signature is a *diagnostic of functional-bass writing*, and the boundary confirms the mechanism. The parity channel's kernel—a_6 equals the even-minus-odd pitch-class parity—is machine-checked in Lean 4 (`ParityA6.lean`, sorry-free, axiom-clean, general even universe). Every ingredient is classical (Rameau/Riemann's functional bass; a_6 = pc-parity, Amiot; the single-note-move rule, Hoffman); the contribution is the **per-voice magnitude law**—a measured, explained, certified, and honestly bounded fact—not new mathematics.

1 Is the bass really the foundation?

When Jean-Philippe Rameau placed the *basse fondamentale* at the centre of harmony in 1722 [1], he made precise an old practical instinct: the lowest voice is not just another melody but the bearer of the chord roots, the ground on which harmony stands. The figured-bass tradition encoded the same priority, and Hugo Riemann's harmonic functions [2] later read tonal motion as a logic of

*Prepared with the assistance of an AI system (Claude, Anthropic). The mathematics is classical and the central finding is empirical; every claim, scope statement, and limitation is the author's responsibility. The boundary of the contribution is stated plainly in Section 8, and the Lean 4 kernel—not the assistant—is what certifies the formalized part (Section 7).

bass-rooted functions. "The bass is the foundation" became one of the most durable claims in music theory—and it stayed a *structural*, qualitative one.

That instinct was, for a century and a half, also a daily practice. In the Baroque *basso continuo*, a single bass line—annotated with figures—was enough for a keyboardist to realize an entire harmony, the upper voices following from the bass rather than the other way round. Rameau's *basse fondamentale* gave the practice a theory: beneath the sounding bass lies a conceptual line of chord *roots*, and harmony is the motion of that line; Riemann later recast the same motion as a compact grammar of tonic, subdominant and dominant functions, again anchored below. Across the common-practice era the bass was thus singled out three times over—in performance, in theory, and in pedagogy—as the voice that *defines* the harmony. The claim was everywhere; a number for it was nowhere.

Two and a half centuries after Rameau, David Lewin [3]—in a two-page note of 1959 that is now seen as the seed of the whole subject—began to turn the internal structure of chords into Fourier coefficients, an idea revived by Ian Quinn as a theory of chord "quality" [4] and built into a discipline by Emmanuel Amiot [5] and Jason Yust [7, 8]. Encode a chord or a passage as a distribution on the twelve pitch classes $\mathbb{Z}_{12}$, take its discrete Fourier transform, and the *magnitudes* $|a_k|$ read as musical **characters**: $|a_5|$ is diatonicity, $|a_4|$ octatonicity, $|a_6|$ whole-toneness, $|a_1|$ chromaticity, and so on. The character profile of a piece can be tracked, charted, even drawn as a pie. But every analysis in this program segments a texture by *time*—sliding windows over the whole-texture pitch-class distribution. **None has asked the orthogonal question: does each voice, by its registral role, carry a different Fourier character profile?**

This note follows that one question. It lets us put Rameau's structural intuition on a measuring instrument: if the bass really is the harmonic foundation, it should leave a measurable spectral *fingerprint*, distinct from the voices above it. It does—but not the fingerprint one would first guess, and only exactly where harmony is functional.

2 Setup: the per-voice DFT

Fix a voice v in a piece and form its **duration-weighted pitch-class distribution** $\mu_v : \mathbb{Z}_{12} \to \mathbb{R}_{\geq 0}$, where $\mu_v(x)$ is the total sounding duration the voice spends on pitch class x (octave collapsed). Its discrete Fourier transform is

$$a_k(v) \;=\; \hat{\mu}_v(k) \;=\; \sum_{x \in \mathbb{Z}_{12}} \mu_v(x)\, e^{-2\pi i\, kx/12}, \qquad k = 0, 1, \ldots, 6,$$

with $a_0 =$ total weight. We compare voices by the **normalized character** $\alpha_k(v) = |a_k(v)|/|a_0(v)| \in [0, 1]$, which is invariant to how loud or long the voice is. The named characters are the standard ones [5, 8]: α_1 chromatic, α_2 quartal, α_3 augmented, α_4 octatonic, α_5 diatonic, α_6 whole-tone. Voices are identified by *register* (mean pitch), so "bass" means the lowest-sounding part, "soprano" the highest—no reliance on part names. The whole construction is linear in the voice decomposition; the contribution here is empirical, in what the measurement reveals.

3 The first guess, and the surprise

P_1. The bass walks by fifths and carries the roots—so is it the most fifth-generated, i.e. the most diatonic *(α_5), voice?*

It is a natural guess, and it is **wrong**. Over the 334 four-part chorales of J.S. Bach (`music21` corpus), the soprano is the most diatonic voice ($\alpha_5 = 0.580$ vs the bass's 0.535, paired $t = -10.4$). The bass is not where diatonicity peaks.

What the measurement *does* reveal is sharper and more robust. The bass is the **spectrally purest** voice: it is the lowest in chromaticity α_1 and the lowest in whole-toneness α_6, by a wide margin (Figure 1, Table is Figure 2). Per-voice means, Bach:

voice	α_1 chrom	α_2	α_3	α_4 oct	α_5 diat	α_6 w-tone
Bass	**0.130**	0.207	0.257	0.118	0.535	**0.079**
Tenor	0.279	0.181	0.261	0.186	0.525	0.190
Alto	0.363	0.233	0.298	0.197	0.505	0.223
Soprano	0.297	0.170	0.233	0.183	**0.580**	0.216

Paired against the soprano: α_1 $t = -20.3$, α_6 $t = -14.8$—the two largest effects in the table. **The bass is the least chromatic and least whole-tone voice; the soprano is the most diatonic; the inner voices are the most chromatic.** One can hear the contrast: the bass of a chorale soloed [♭ audio] against its alto [♭ audio].

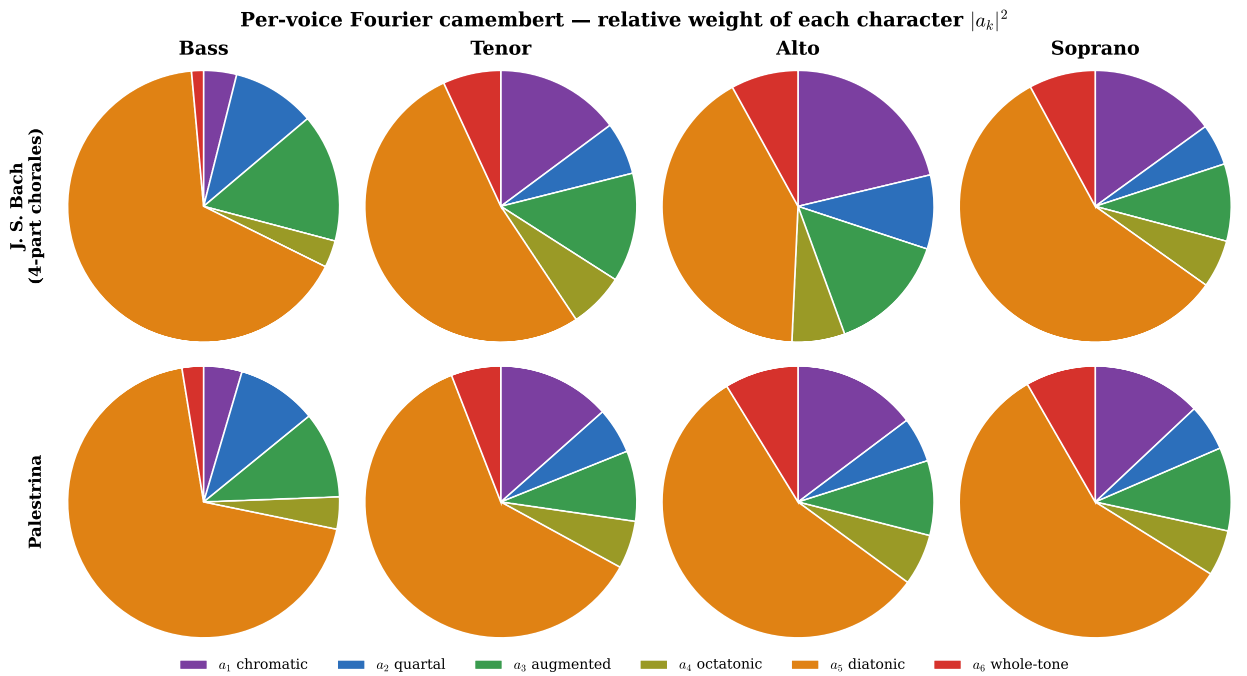

Figure 1: The per-voice Fourier "camembert": the relative weight of each character ($|a_k|^2$, Parseval) for Bass/Tenor/Alto/Soprano, in Bach chorales (top) and Palestrina (bottom). The **bass pie** is visibly thin in the chromatic (purple) and whole-tone (red) slices and fat in diatonic (orange); the **alto** carries the largest chromatic wedge. The signature is legible at a glance and reproduces across two corpora two centuries apart.

Not a cardinality artifact. A spread-out distribution has small high-k coefficients, so one might suspect the bass simply uses fewer pitch classes. The opposite is true: the bass uses *more* distinct pitch classes than the soprano (9.5 vs 7.5 on average) and has higher entropy, yet has the lowest α_1. The low chromaticity is genuine distributional structure, not a counting effect.

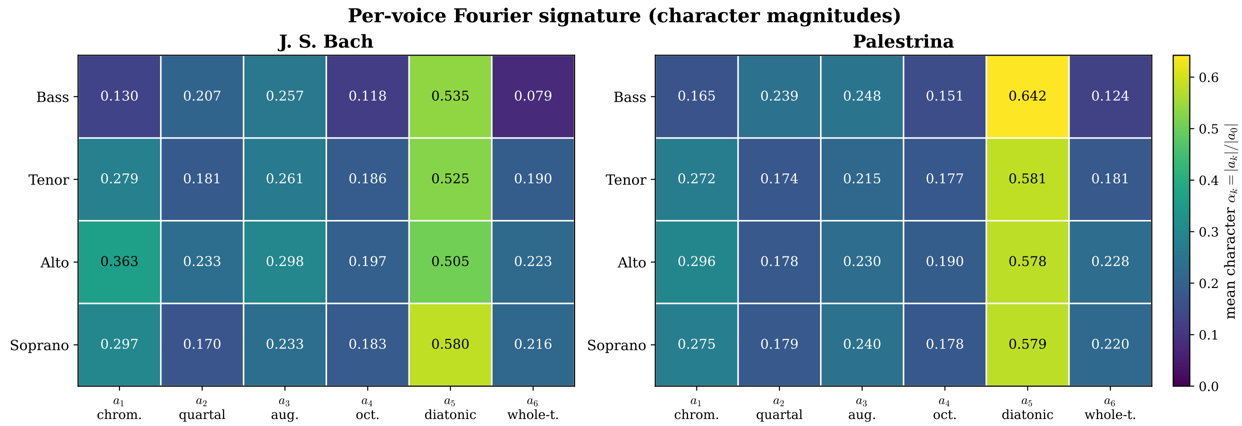

Figure 2: The whole signature as a matrix: voices $\times$ characters α_k, Bach and Palestrina. The **bass row** is darkest in a_1 and a_6 and brightest in a_5, in both corpora; the **alto** is brightest in a_1. The diatonic column dominates every voice (tonal/modal music), but the *contrast between voices* lives in a_1 and a_6.

4 Why: root realization, in two channels

P_2. Why is the bass spectrally pure—and why in a_1 and a_6 specifically?

Because, to 95% cosine fidelity (duration-weighted, Bach), the bass distribution *is* the chord-root distribution: the bass realizes the harmonic roots. Functional root motion proceeds by fifths and fourths—intervals 7 and 5, both **odd**. This single fact drives two distinct spectral channels.

The parity channel (a_6). The whole-tone coefficient is, exactly, $a_6 = \sum_x \mu(x)\,(-1)^x =$ (even-pc weight) − (odd-pc weight): it reads the even/odd **parity imbalance** of the distribution (this identity is the certified kernel of Section 7). An odd interval flips pitch-class parity; a root progression that walks by fifths therefore alternates parity and *balances* the two whole-tone collections, sending $a_6 \to 0$ [♭ audio]. Empirically, the correlation of α_6 with the voice's odd-interval fraction is -0.49: the more a voice moves by odd intervals (the bass most of all), the lower its whole-toneness.

The dispersion channel (a_1). The coefficient a_1 measures chromatic *clustering*—it is large when weight piles up in a contiguous arc of the chromatic circle. The functional bass leaps; it spans the widest tessitura of the four voices (≈ 18 semitones vs ≈ 12), so its pitch-class weight is spread widely around the circle rather than clustered, and α_1 is low [♭ audio]. Empirically, the correlation of α_1 with voice range is -0.67. Figure 3 shows this directly: the bass's weight fans out around the clock, the alto's bunches into an arc.

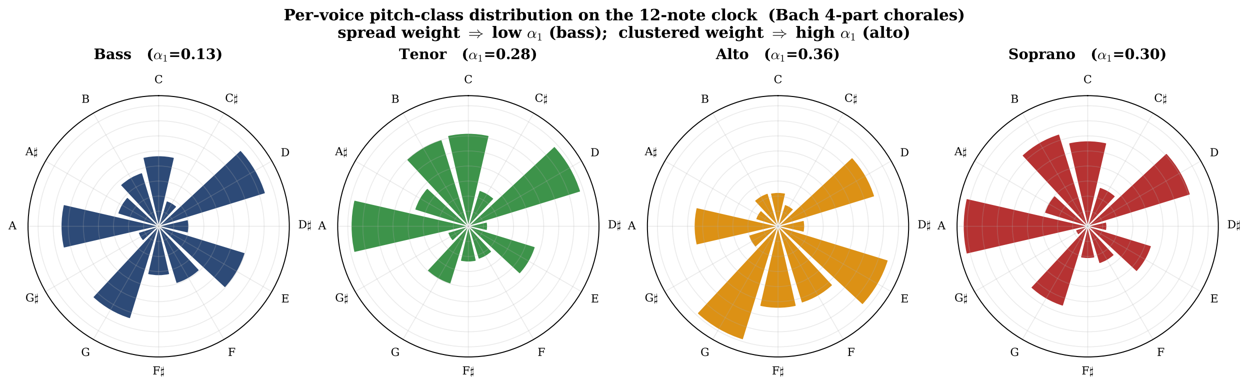

Figure 3: The per-voice pitch-class distribution drawn on the twelve-note clock (Bach). The **bass** ($\alpha_1 = 0.13$) spreads its weight widely around the circle (low chromatic clustering); the **alto** ($\alpha_1 = 0.36$) bunches into a narrow arc (high clustering). Spread weight ⇒ low a_1; clustered weight ⇒ high a_1—the dispersion channel, seen.

A mechanism that was *not* the cause. It is tempting to attribute the bass's low chromaticity to its melodic motion—"the bass avoids semitone steps." It does not: the correlation of α_1 with the fraction of semitone melodic steps is just +0.01, and the bass has as many semitone steps as the other voices. The cause is in the pitch-class *distribution* (which classes, how spread), not in the melodic surface. We record the refuted mechanism because it sharpens the true one.

5 The historical sweep: a diagnostic of functional-bass texture

P_3. If the cause is the functional bass, the law should appear exactly where the bass bears roots—and fade where it does not. Does it?

It does—and the cleanest way to see it is to walk four corpora across four centuries, each embodying a different relationship between the lowest voice and the harmony. In the Italian **Trecento** (the fourteenth-century *Ars Nova* of Landini and his contemporaries) polyphony is built line by line above a structural *tenor*; the lowest sounding voice is not yet a bearer of harmonic roots, and the signature is simply absent (α_1 bass-purer-than-top in 52% of works, α_6 in 58%—no better than the 50% coin toss). By Palestrina's **stile antico** (sixteenth-century modal counterpoint, the music of the Counter-Reformation) a clear lowest voice already governs the harmonic implication of each sonority, and the signature is strong (87%, 77%). It is strong again two centuries later in **Bach**'s chorales (85%, 73%)—the apotheosis of functional four-part harmony, where the bass simply *is* the figured-bass foundation of §1. The telling case is **Monteverdi**: his madrigals *postdate* Palestrina, yet show the *weaker* effect (65%, 53%—near chance), because the expressive *seconda prattica* madrigal is imitative and text-driven, scattering its chromatic colour across five near-equal voices in which no single line is the privileged root-bearer. So the signature tracks **texture, not date** (Figure 4): it is strong exactly where the bass realizes roots and fades where it does not. The boundary is not a failure of the law but its confirmation—the same root-realization mechanism that predicts the effect predicts where it must vanish—and it recasts the effect as a **diagnostic of functional-bass writing**, a spectral fingerprint of the very practice Rameau theorized.

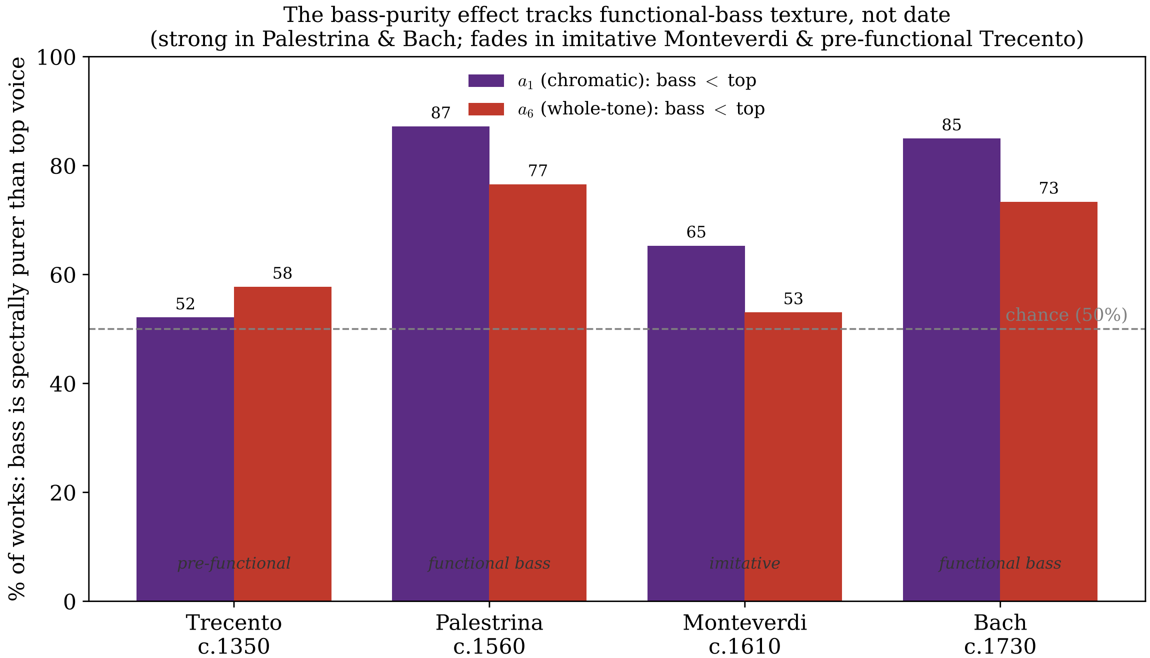

Figure 4: The bass-purity effect across four corpora. Bars: percentage of works in which the bass is spectrally purer than the top voice, in a_1 (chromatic) and a_6 (whole-tone), against the 50% chance line. Strong in functional-bass homophony (Palestrina, Bach), it fades in imitative (Monteverdi) and pre-functional (Trecento) textures—tracking the role of the bass, not chronology.

6 The dual: inner voices carry the colour

P_4. If the bass is the purest, which voice is the least pure—and why?

The chromaticity ordering is Bass 0.130 < Tenor 0.279 < Soprano 0.297 < **Alto** 0.363: the inner voices, the alto above all, carry the most chromatic colour [♭audio]. This is the dispersion channel run in reverse: inner voices occupy the *narrowest* tessitura ($\approx$ 11–12 semitones), so their weight is compressed into a chromatic band and α_1 is high. It is the registral-spread story, not a melodic-chromaticism one (the semitone-step correlation is null, as in §4). The four voices sit, in the diatonic-phase plane (Φ_3, Φ_5), almost on top of one another (Figure 5)—they share a key—so the inter-voice information is in the *magnitudes* α_1, α_6, not the phases: a bass and a soprano differ in spectral *purity*, not in tonal centre.

This inverts the textbook image in a precise way. The *outer* voices are the structural frame—the bass rooting the harmony, the soprano carrying the tune—while the *inner* voices are where the chromatic colour lives: the home of suspensions, passing and neighbour tones, and the small chromatic voice-leading that fills out a four-part texture. That the alto is the single most chromatic voice is thus the spectral echo of a practice as old as the *altus* of Renaissance polyphony—the inner parts move by the short, often chromatic steps that the bass, leaping by fifths, never takes, and the soprano, singing the diatonic melody, mostly avoids. Purity below, tune above, colour within: the Fourier characters recover the division of labour of the four-voice chorale itself.

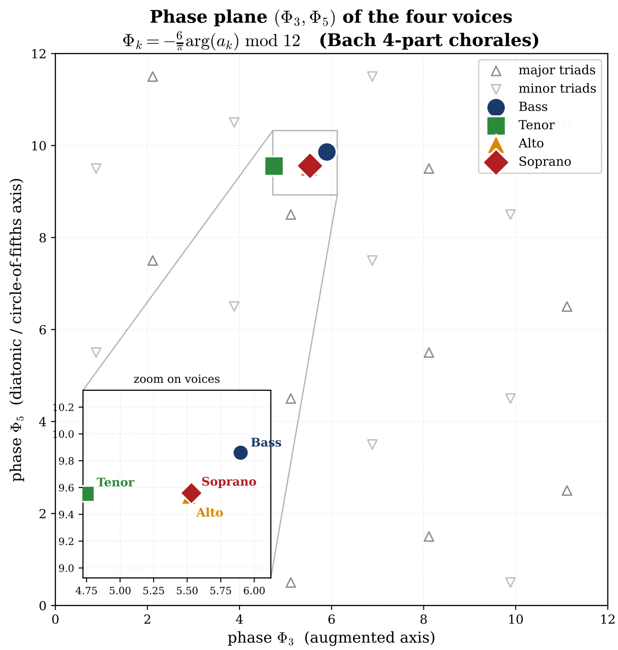

Figure 5: The four voices in the diatonic phase plane (Φ_3, Φ_5) against the 24 major/minor triads. The voices are near-coincident (a shared key): the per-voice contrast is carried by the character *magnitudes* (a_1, a_6), not the phases.

7 What is machine-checked

P_5. The empirical law rests on a measurement; what part of it is certified?

The parity channel's kernel—the identity that makes the whole-tone coefficient a parity count—is proved in Lean 4 (`ParityA6.lean`, namespace `ParityA6`, sorry-free; built on the project's `Fourier.lean`). Table 1 lists the results. The headline is that a_6 *is* the even-minus-odd parity imbalance, with the order-2 frequency reading parity in *any* even universe $\mathbb{Z}_{2m}$, not only $\mathbb{Z}_{12}$. The axiom audit (`#print axioms`) returns at most `[propext, Classical.choice, Quot.sound]` on every named result—no `sorryAx`, no `Lean.ofReduceBool`—confirmed by recompiling in the project's fast-loop environment (EXIT 0).

Lean result (`ParityA6.lean`)	**Statement**
`Ahat_six_eq_sum_neg_one_pow`	$\hat{A}(6) = \sum_{x \in A} (-1)^x$ — the whole-tone coefficient is the signed parity sum.
`Ahat_six_eq_card_sub`	$\hat{A}(6) = \#(\text{even pcs}) - \#(\text{odd pcs})$ — the parity imbalance, over $\mathbb{C}$.
`Ahat_six_eq_zero_iff`	$\hat{A}(6) = 0 \iff \#(\text{even}) = \#(\text{odd})$ — a_6 vanishes exactly at even/odd balance.
`dftN_half_eq_sum_neg_one_pow`	General even universe: in $\mathbb{Z}_{2m}$, the order-2 frequency reads parity, $\hat{\mu}(m) = \sum_x (-1)^x$.

Table 1: The parity kernel of the a_6 channel, machine-checked. Four `example` witnesses accompany it (C major $a_6 = 1$; whole-tone $a_6 = 6$; diatonic $a_6 = -1$; a balanced tetrachord $a_6 = 0$). The math is classical (see §8); this is its first formalization.

8 Novelty, scope, and what this note does not claim

Contribution, stated plainly. This is an **empirical** note with a **formalized kernel**, not new mathematics. Every explanatory ingredient is classical: the functional bass is Rameau [1] and Riemann [2]; the DFT-of-pitch-class-distributions program and the character readings ($a_5 =$ diatonicity, etc.) are Lewin [3], Quinn [4], Amiot [5], Yust [7, 8]; the identity $a_6 =$ pitch-class parity is Amiot (*The Torii of Phases* [6], "a_6 takes only two values, depending on the number of odd pitches"), and the single-note-move mechanism behind it is Hoffman [10]. **The new content is the per-voice magnitude law**: that, in functional-bass textures, the bass is the spectrally purest voice (lowest α_1, α_6), measured across corpora, explained by root realization in two channels, and bounded by texture. The DFT program reads textures by *time*; the per-voice (registral) magnitude contrast is the orthogonal cut, and it is, to our search, unstated.

What this note does *not* claim.

- **Not a theorem.** The law is an empirical regularity; its mathematical kernel is the linearity of the DFT over the voice decomposition. Only the parity identity (Table 1) is a Lean theorem.
- **Not universal.** The signature is a diagnostic of *functional-bass* texture: strong in Palestrina and Bach, weak/absent in Monteverdi madrigals and the Trecento (§5).
- **Not the diatonicity guess.** The bass is *not* the most diatonic (a_5) voice; that is the soprano. The invariant is the bass's low a_1 and a_6, not high a_5.
- **Not a melodic claim.** The low chromaticity is a property of the pitch-class *distribution* (dispersion / parity), not of semitone melodic motion (correlation +0.01).
- **Not priority over the DFT program.** Lewin, Quinn, Amiot, and Yust each precede this; we cite, measure, formalize, and do not supplant. The parity identity is Amiot/Hoffman; our `ParityA6.lean` is its first formalization, and it is *distinct* from—not a corollary of—Amiot's tritone lemma (which concerns the *odd*-index coefficients).

Open frontier. A larger corpus and more styles would map the texture boundary precisely (where, quantitatively, does "functional-bass" begin?); a per-voice study of the *phases* (rather than magnitudes) is untouched here; and the dispersion channel invites a clean statement bounding a_1 by a voice's circular spread.

Data and code availability

The per-voice measurements (`music21`: 334 Bach four-part chorales, 367 Palestrina works, plus the Monteverdi and Trecento corpora), the figure toolkit (`voice_figs.py`, regenerating Figures 1–5 from the corpora), the sonifications (`media/`), and the Lean source `ParityA6.lean` accompany this note; each Lean result is checkable by `#print axioms` and each script is re-runnable.

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